

LIQUEO

Knowledge Discovery & Reuse System

Professional User Guide & System Documentation

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TABLE OF CONTENTS

1. System Overview & Key Features

2. User Interface Guide

2.1 Add Document Tab

2.2 Search Tab

2.3 Recommendations Tab

2.4 Knowledge Workflow Tab

3. How to Use Each Feature

4. 9-Step Workflow Explained

5. Technical Architecture

6. Quick Start Guide

7. Best Practices & Tips

8. Troubleshooting & FAQs

1. SYSTEM OVERVIEW & KEY FEATURES

Liqueo is an intelligent knowledge discovery and reuse system that helps financial and business consultants discover, analyze, and adapt proven consulting approaches from past engagements. Instead of reinventing methodologies for each new project, consultants use Liqueo to find similar past work, understand what succeeded, and adapt those approaches for new contexts.

Core Features

Semantic Search: AI-powered search finds conceptually similar past engagements, not just keyword matches

Pattern Analysis: System automatically extracts patterns and best practices from similar cases

LLM Synthesis: AI generates structured recommendations based on successful approaches

9-Step Workflow: Guided process from problem identification through knowledge storage

Multi-format Upload: Import PDF, Word, Excel, CSV, and text documents

Rich Metadata: Comprehensive tagging for industry, engagement type, value, duration

Source Traceability: All recommendations include links back to original engagements

No API Required: System works with or without cloud APIs (graceful degradation)

Business Value Delivered

Metric	Without Liqueo	With Liqueo	Improvement
Proposal Time	70 hours (3-5 days)	70 minutes	60x faster
Content Reuse	0-20%	70-80%	80% from precedents
Quality Consistency	Varies by consultant	Based on proven methods	Standardized excellence
Knowledge Loss	High (staff turnover)	Systematic capture	Preserved for future

2. USER INTERFACE GUIDE

2.1 Add Document Tab - Upload & Management

The Add Document tab is where you capture new consulting engagements and add them to the knowledge base. This is the primary way Liqueo learns and builds institutional knowledge over time.

What You See:

- File upload area (drag & drop or click to browse)
- Form fields for engagement metadata
- Industry dropdown selector
- Fields for value, duration, team structure
- Text areas for approach description and outcomes
- Save button and confirmation messages

Step-by-Step Usage:

1. Click 'Add Document' tab at the top of the page
2. Drag a document (PDF/Word/Excel) into the upload area or click to browse
3. System automatically extracts text from your document
4. Fill in the metadata fields (Industry, Value, Duration, Client Name)
5. Enter the consulting approach used (2-3 paragraphs)
6. Describe key outcomes and measurable results
7. Add tags for better searchability (comma-separated)
8. Review the preview of your submission
9. Click 'Save to Knowledge Base'
10. Confirmation message appears - engagement is now discoverable

2.2 Search Tab - Finding Similar Engagements

The Search tab uses semantic search to find past engagements conceptually similar to your current challenge. Unlike keyword search, semantic search understands meaning - a query for "banking cost reduction" will find results about "financial services optimization" if the engagement is relevant.

What You See:

- Large search input field
- Optional filter dropdowns (Industry, Type)
- Search button
- Results displayed as cards with relevance scores
- Each result shows: Title, Industry, Value, Duration, Relevance %
- 'View' button to see full engagement details
- Modal popup with complete engagement information

How to Search:

1. Click 'Search' tab
2. Type your challenge description (e.g., 'fintech infrastructure optimization cost reduction')
3. Optionally select Industry or Type filters
4. Click 'Search'
5. Results appear ranked by relevance score (0-100%)
6. Click 'View' on any result to see full details in a modal
7. Read the consulting approach, outcomes, and team structure
8. Close modal to continue browsing results

2.3 Recommendations Tab - AI-Powered Analysis

The Recommendations tab uses AI to analyze multiple similar cases and synthesize best practices. It generates structured recommendations including proven approaches, success factors, and typical challenges.

What You Get:

Recommended Approach: A structured methodology synthesized from successful cases

Success Factors: Key elements that made similar engagements successful

Typical Challenges: Common obstacles and how they were overcome

Team Composition: Recommended roles and team structure

Timeline Estimate: Typical project duration for similar engagements

Confidence Score: How confident the AI is in its recommendation (based on number of similar cases)

Source References: Links to the specific engagements analyzed

2.4 Knowledge Workflow Tab - 9-Step Guided Process

The Knowledge Workflow tab is the core of Liqueo. It guides you through all 9 steps of the knowledge reuse process. Each step is designed to maximize the value you get from past engagements when creating new proposals.

The 9 Steps:

Step 1: Identify Problem

Describe your new engagement challenge with context (industry, client type, specific requirements, constraints)

Step 2: Search Knowledge

Liqueo searches the knowledge base for similar past engagements using semantic search

Step 3: Identify Related Documents

System extracts relevant documentation: templates, lessons learned, success patterns, team structures

Step 4: AI Summarize

LLM analyzes the similar cases and generates executive summary, recommended approach, outcomes, and recommendations

Step 5: Review & Evaluate

You review the AI analysis and similar cases. Take notes on applicability to your new context.

Step 6: Select Content

Choose which consulting approach, timeline, team structure, and methodologies to reuse from similar engagements

Step 7: Create New Output

System creates new engagement using auto-populated template based on your selected similar case

Step 8: Tag & Classify

Add metadata (industry, engagement type, tags) to your new engagement for future discovery

Step 9: Store Knowledge

Save to knowledge base. Your new engagement is now discoverable for the next consultant facing a similar challenge.

3. HOW TO USE EACH FEATURE

Real-World Scenario: Fintech Proposal

Situation: Friday 4pm. A fintech client sends an RFP for core banking infrastructure optimization with 30% cost reduction requirement. Proposal due Monday 9am. You have 70 hours to deliver.

Without Liqueo: You start from scratch. Find relevant templates in email. Piece together an approach from memory. Result: 50-70 hours of work, inconsistent with past successes.

With Liqueo: Follow this process in 70 minutes...

1. Go to Recommendations tab
2. Search: 'fintech core banking optimization cost reduction'
3. System returns 3 similar banking cases ranked by relevance
4. Click 'View' on Investment Bank case (78% relevant, \$2.8M value, 40% achieved)
5. Read their 3-phase approach: Assess → Design → Implement
6. Examine their team structure and timeline (12 months for \$2.8M project)
7. Go to Recommendations tab - see AI synthesis showing patterns across 3 cases
8. Go to Knowledge Workflow tab, follow all 9 steps
9. Step 7 auto-populates your proposal based on Investment Bank template
10. Customize for your fintech client's specific requirements
11. Add tags and store - it's now ready for next consultant facing similar challenge

Result: Proposal completed in 70 minutes using proven 3-phase approach from similar case. Quality based on successful precedent. Ready to submit.

4. TECHNICAL ARCHITECTURE

Component	Description
Web UI (Streamlit)	4-tab user interface: Add, Search, Recommendations, Workflow
Semantic Search	AI embeddings for meaning-based search (OpenAI/Anthropic)
Recommendation Engine	Pattern matching across similar engagements
LLM Synthesis	Claude/GPT analysis for best practice generation
Knowledge Base	JSON filesystem storage of all engagements
Workflow Engine	Tracks 9-step process and auto-populates templates

Operating Modes

- **Full Mode:** OpenAI or Anthropic APIs available. Complete feature set: semantic search + LLM synthesis.
- **Hybrid Mode:** One API selected (your choice). Can do embeddings OR synthesis.
- **Manual Mode:** No APIs needed. Uses keyword search. Perfect for demos and testing.

5. QUICK START GUIDE (5 MINUTES)

```
$ git clone https://github.com/hemalp143/Liqueo.git
$ cd Liqueo
$ python3 -m venv venv
$ source venv/bin/activate
$ pip install -r requirements.txt
$ python load_sample_data.py
$ streamlit run app.py
```

First Steps After Installation

- ✓ Sample data pre-loaded (3 banking engagements)
- ✓ Try Search tab: 'banking cost optimization'
- ✓ Go to Recommendations to see AI analysis
- ✓ Follow Knowledge Workflow for complete 9-step experience
- ✓ Upload your first engagement in Add Document

6. BEST PRACTICES & TIPS

For Adding Documents

- ✓ Use consistent industry names (Financial Services, not 'Bank', not 'Finance')
- ✓ Be specific about engagement type (M&A, Cost Optimization, Process Improvement, Digital Transformation)
- ✓ Include measurable outcomes (% cost reduction, revenue increase, timeline improvement)
- ✓ Write clear consulting approach (3-5 key phases or steps)
- ✓ Add meaningful tags that future consultants might search for
- ✓ Include team size and key roles

For Effective Searching

- ✓ Be specific about your challenge (not just 'banking', but 'banking cost optimization')
- ✓ Include measurable goals if known (e.g., '30% cost reduction')
- ✓ Describe industry and company type for better matches
- ✓ Try multiple search queries to see different angles
- ✓ Use filters to narrow results if too many returned
- ✓ Review top 3 results - patterns will emerge

For Workflow Success

- ✓ Don't skip Step 5 - really evaluate applicability to your context
- ✓ In Step 6, select multiple elements from different cases if helpful
- ✓ Step 7 creates templates - customize heavily for your specific client
- ✓ Spend time on Step 8 tagging - makes your engagement discoverable for future use
- ✓ The goal is 70-80% reuse from past engagements, with customization

7. TROUBLESHOOTING & FAQs

Q: Do I need to provide API keys?

A: No. Liqueo works without API keys using keyword search. Keys are optional for semantic search + AI synthesis.

Q: Why am I getting few search results?

A: Knowledge base may be small. Start by uploading several engagements. Liqueo improves with more data.

Q: Can I upload PowerPoint files?

A: Not yet. PowerPoint support is coming in Phase 4. Use PDF or Word documents for now.

Q: How long should documents be?

A: Any length works, but 3-10 pages is typical. Focus on consulting approach and measurable outcomes.

Q: What if Liqueo suggests an irrelevant case?

A: Just skip it. Go to the next result. Similarity scores show relevance (aim for >70%).

Q: Can I delete or edit engagements?

A: Currently read-only. Deletion feature comes in Phase 2. Edit by uploading a new version.

Q: How is data stored?

A: In a .liqueo/ directory as JSON files. No database required. Can be backed up as a folder.

Q: Can multiple users access the same knowledge base?

A: Phase 1 is single-user. Multi-user support with role-based access comes in Phase 5.

